

Repaired Native-VC Brill Authority Qualification

AthenaK evidence campaign

24 August 2026

NATIVE_AMR_UNSTABLE

Abstract

We generated a fresh N256 vertex-centered Cartoon AMR authority after repair of derefinement-slot corruption, verified deterministic record/replay, and replayed the authority at N128 and N512. The repaired scheme converges cleanly through central proper time $\tau \simeq 4$. A later N256 extension develops large constraints on a fixed 44-leaf hierarchy before native refinement activates, then undergoes a refinement cascade to logical level 23 and stalls near $t = 11.192887945M$. The result qualifies the early window but is neither a full Figure-3 reproduction nor a convergence result for late collapse.

1 Authority and deterministic replay

The reviewed numerical baseline is commit 6dd20656a305f2543bbbd7001550c6ac67019180. A diagnostics-only repair at d63519328214a6315a9cc1f7d5e4a1aa4bca21b0 replaces unordered CUDA atomics in regional history sums with a Kokkos reduction. It changes no evolution variable, AMR decision, transfer operator, equation, gauge, RK, KO, or CFL choice. In the post-repair aggregate host run, 140/141 enabled tests passed; the sole AMR-history timeout passed in isolation in 177.96 seconds. The focused one-A100 suite passed 20/20 tests.

N256 record and replay have identical accepted events, times, checksums, history bytes, auxiliary diagnostics, and numerical payloads for all 33 binary outputs and six restarts. Complete binary/restart containers differ in their authenticated parameter headers because one records mode **record** and the other **replay**; numerical identity follows the repository contract and hashes bytes after **<par_end>**.

The repaired and historical authorities agree exactly through event 3. The historical tree immediately re-refines at $t = 0.30305969M$ to 74 leaves. The repaired tree instead stays at 44 leaves through $t = 6.5M$ and next refines only at $t = 10.27866870M$ (Figure 1).

2 Qualified early and $\tau \simeq 4$ windows

N128, N256, and N512 replay the same logical tree at identical physical event times. At $t = 6.5M$, their central proper times are 3.98615, 3.98694, and 3.98712. Every global constraint decreases monotonically with resolution.

Table 1: Global constraint diagnostics at $t = 6.5M$.

Resolution	C	H	M	Z
N128	5.5801×10^{-2}	2.1558×10^{-2}	8.6428×10^{-3}	4.0712×10^{-3}
N256	3.2850×10^{-3}	3.5072×10^{-4}	4.2914×10^{-4}	5.0544×10^{-4}
N512	1.8080×10^{-3}	1.2559×10^{-4}	5.7452×10^{-5}	4.0103×10^{-4}

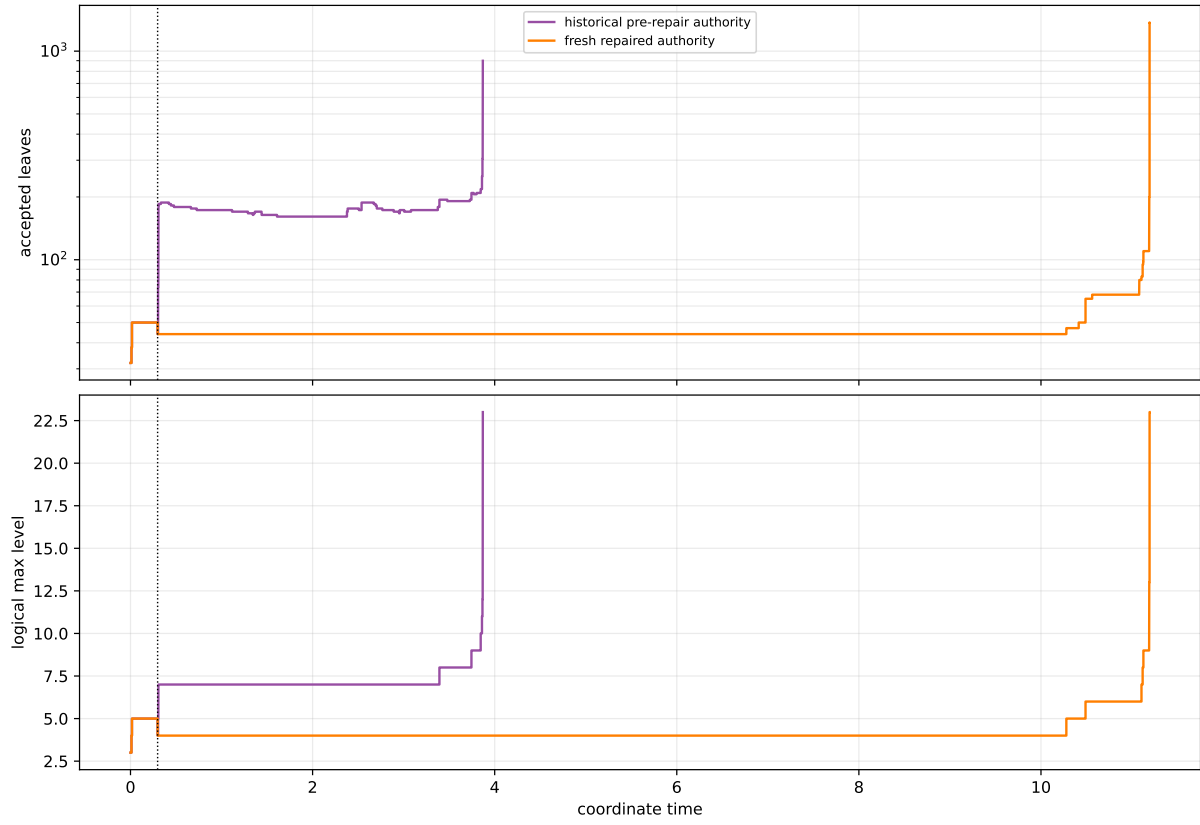


Figure 1: Historical pre-repair and fresh repaired hierarchy histories. The dotted line marks the repaired event-3 derefinement.

Median effective history orders are 6.746, 6.794, 6.127, and 7.029 for C, H, M, Z . Every nonzero evolved field has positive minimum trusted-core order; the lowest is 1.053 for Θ . Minimum χ is 0.3211, the minimum conformal-metric SPD pivot is 0.1974, and sampled same-level and coarse-fine shared-node spreads are zero.

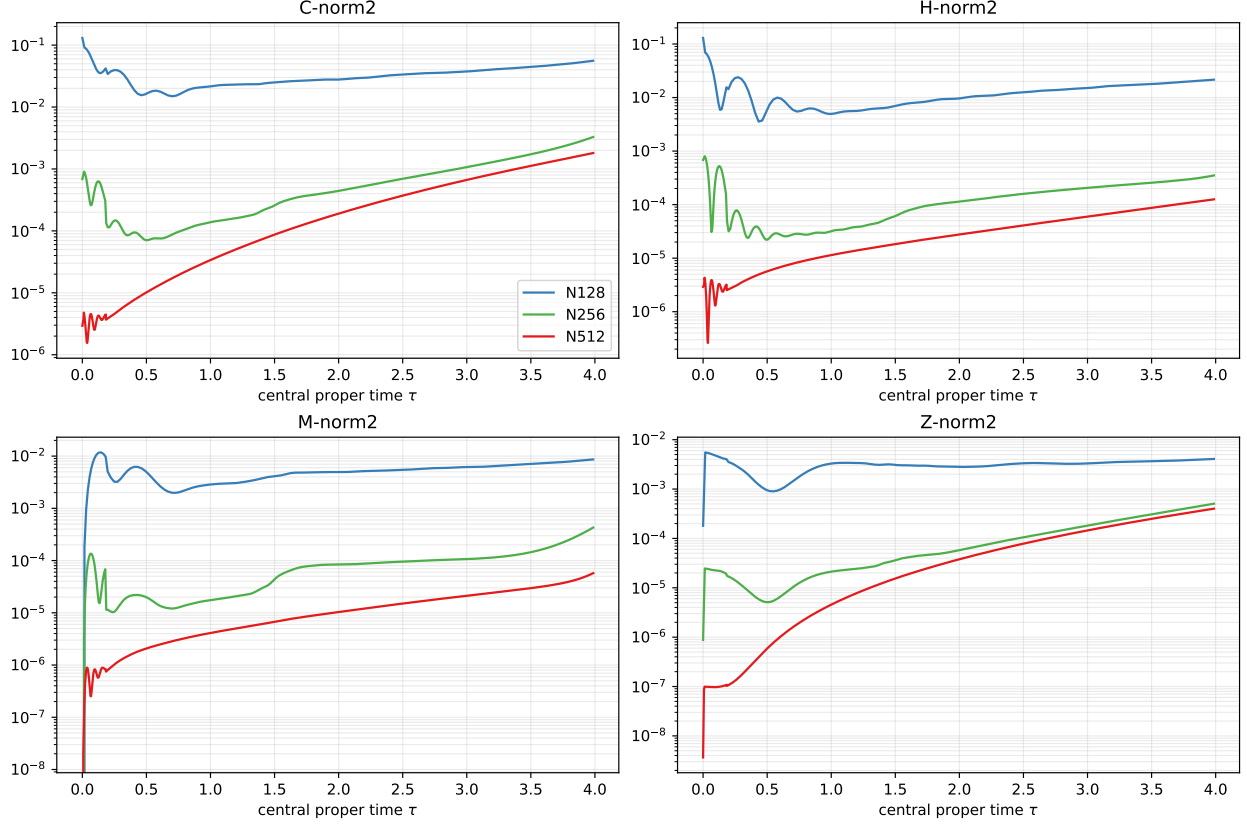


Figure 2: Global constraints through the qualified $\tau \simeq 4$ milestone.

3 Native-AMR failure before $\tau \simeq 7$

The later N256 segment first exceeds $C = 0.01$ at $t = 7.2894893M$ ($\tau = 4.45109$) and $C = 1$ at $t = 9.2042653M$ ($\tau = 5.61265$), while the hierarchy remains fixed at 44 leaves and physical level 1. The first late refinement occurs only at $t = 10.2786687M$ ($\tau \simeq 6.28$), when the nearest history sample already has $C \simeq 20.9$.

The hierarchy then reaches logical level 7 at $t = 11.1038229M$. Event 22 at $t = 11.19248494M$ reaches logical level 14 and begins an effectively coordinate-time-stagnant cascade. At cancellation, event 159 has 1,367 leaves, logical level 23, physical refinement level 20, and $\Delta t = 1.07 \times 10^{-8}M$. The global C diagnostic is 1.40×10^{14} and the domain maximum absolute Kretschmann scalar is 1.60×10^{25} , while the sampled axis-center value is only 7.40×10^{-3} . The runaway is therefore noncentral in this diagnostic.

The failed authority contains 160 accepted hierarchy states and 159 topology transactions: 144 grow, 13 shrink, and two preserve the leaf inventory. It records 1,521 created leaves, 186 deleted leaves, 888 balance-induced leaves, and 21 sign changes between successive leaf-count increments. The inventory peaks at 1,373 leaves and terminates at 1,367. Refinement and derefinement requests coexist in 146 event records; these are per-MeshBlock diagnostic flags, not accepted-coarsening counts.

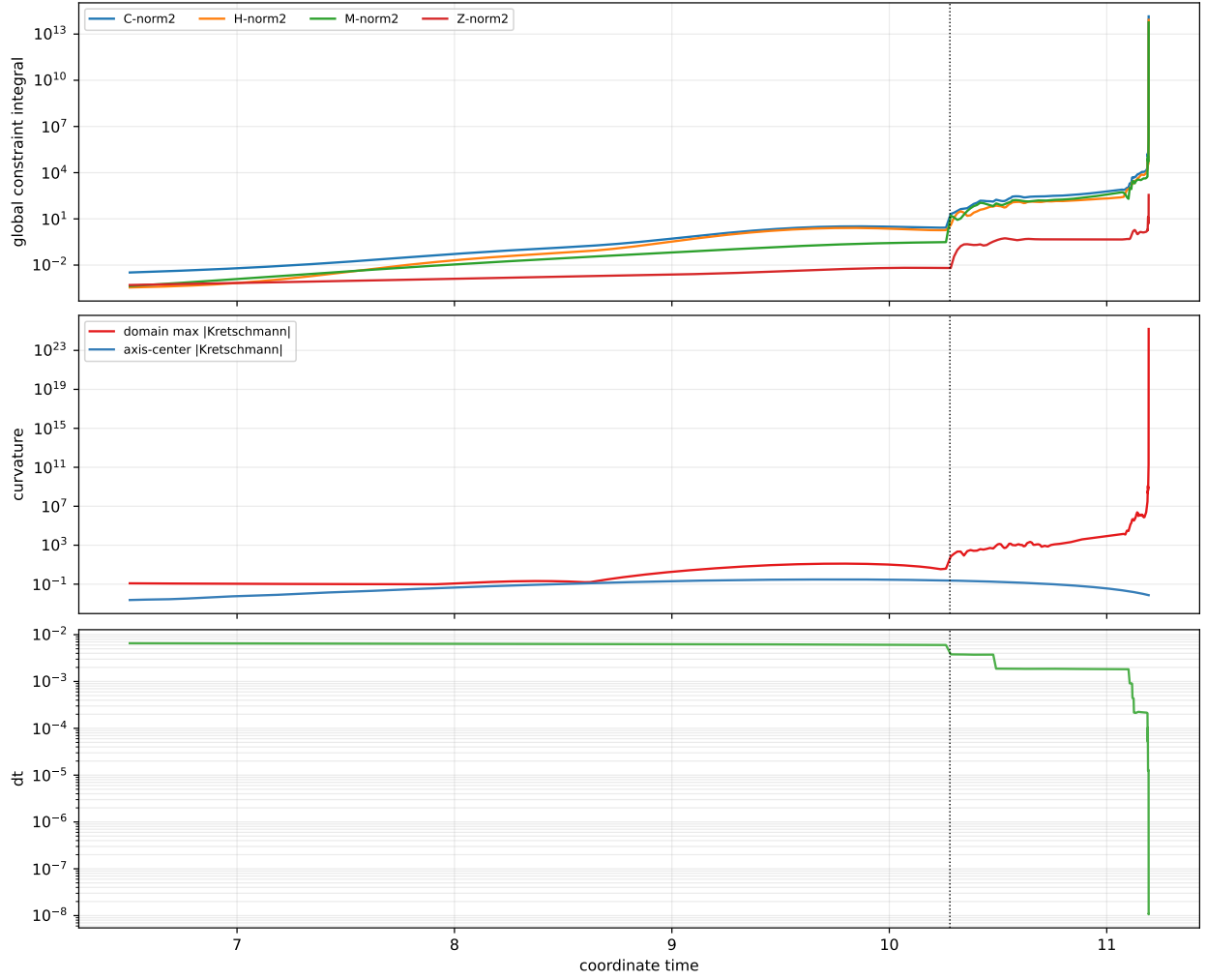


Figure 3: Constraint growth, noncentral curvature runaway, and timestep collapse in the failed N256 extension. The dotted line marks the first late accepted refinement.

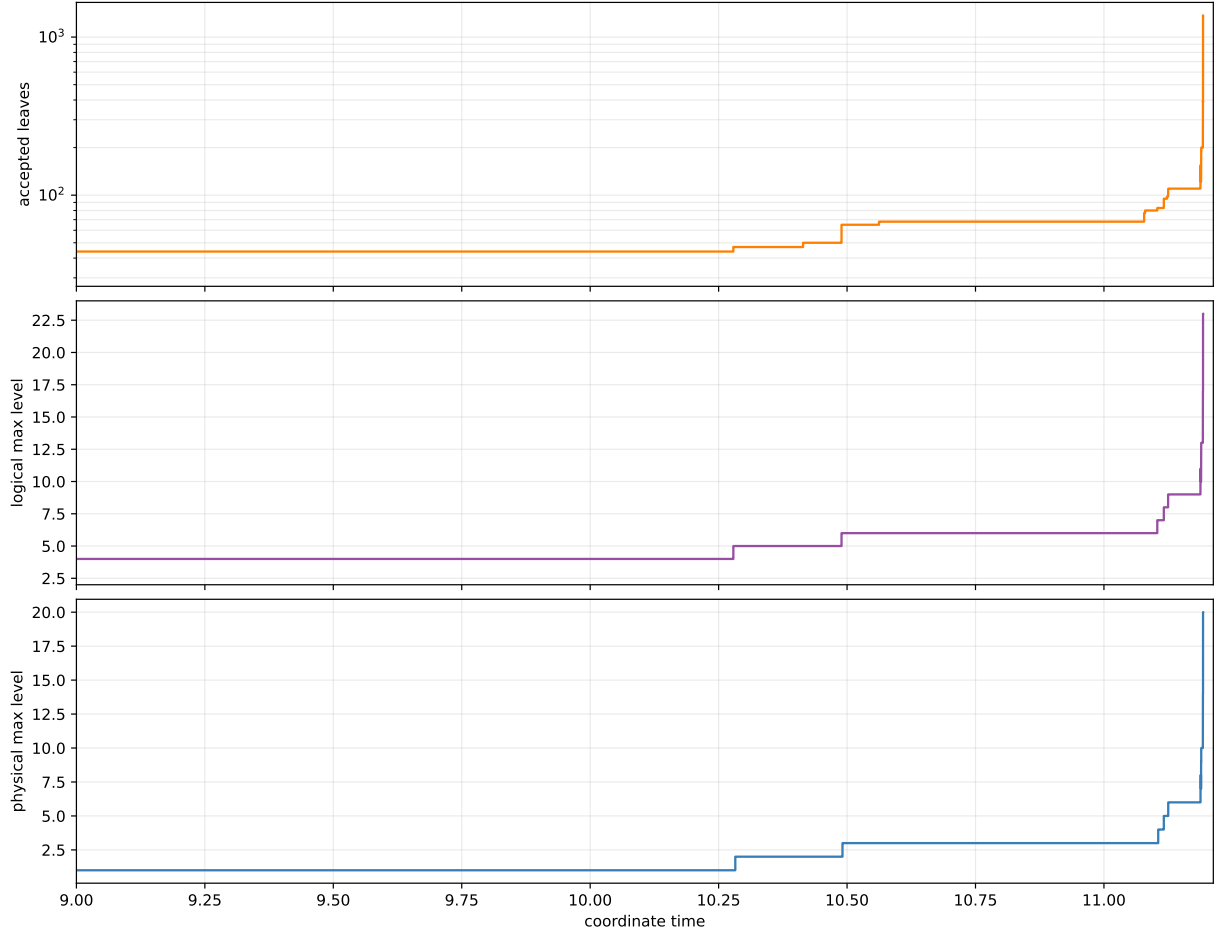


Figure 4: Accepted leaves and logical/physical refinement levels during the late native-AMR cascade.

4 Interpretation and qualification boundary

Observation: substantial constraint growth begins on the fixed parent hierarchy before the refinement cascade. Inference: the current chi-only sensor activates late relative to the shortest or error-bearing scale, or a bulk numerical mode grows before refinement exposes it. Hypothesis: repeated high-order transfer and interface coupling then accelerates the existing error. This campaign does not distinguish those mechanisms and does not establish a source-level bug.

Job 57525753 was cancelled at the fail gate. No N128/N512 replay was launched for the failed authority, and no $\tau \simeq 10.5$ or full published-interval calculation was attempted. Figure 5 is only a direct, unshifted overlay over the qualified partial interval.

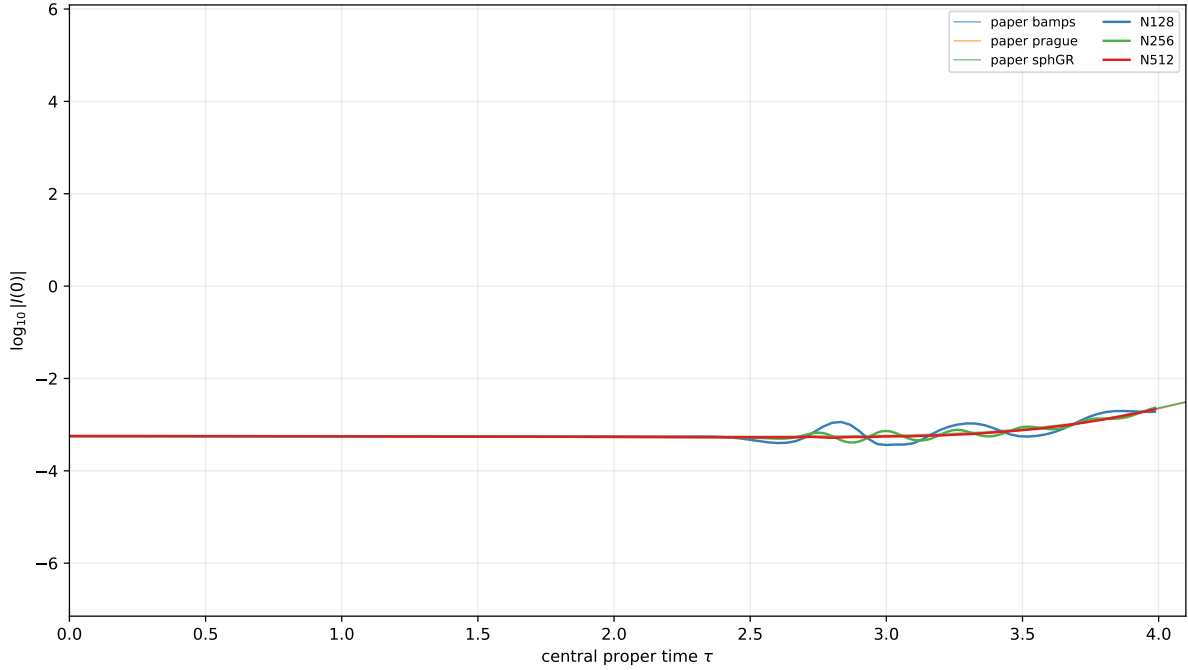


Figure 5: Authenticated Figure-3 curves and AthenaK results over the qualified partial interval. No time shift, amplitude scaling, or coordinate rescaling is applied.